Vibe Coding with GitHub Copilot

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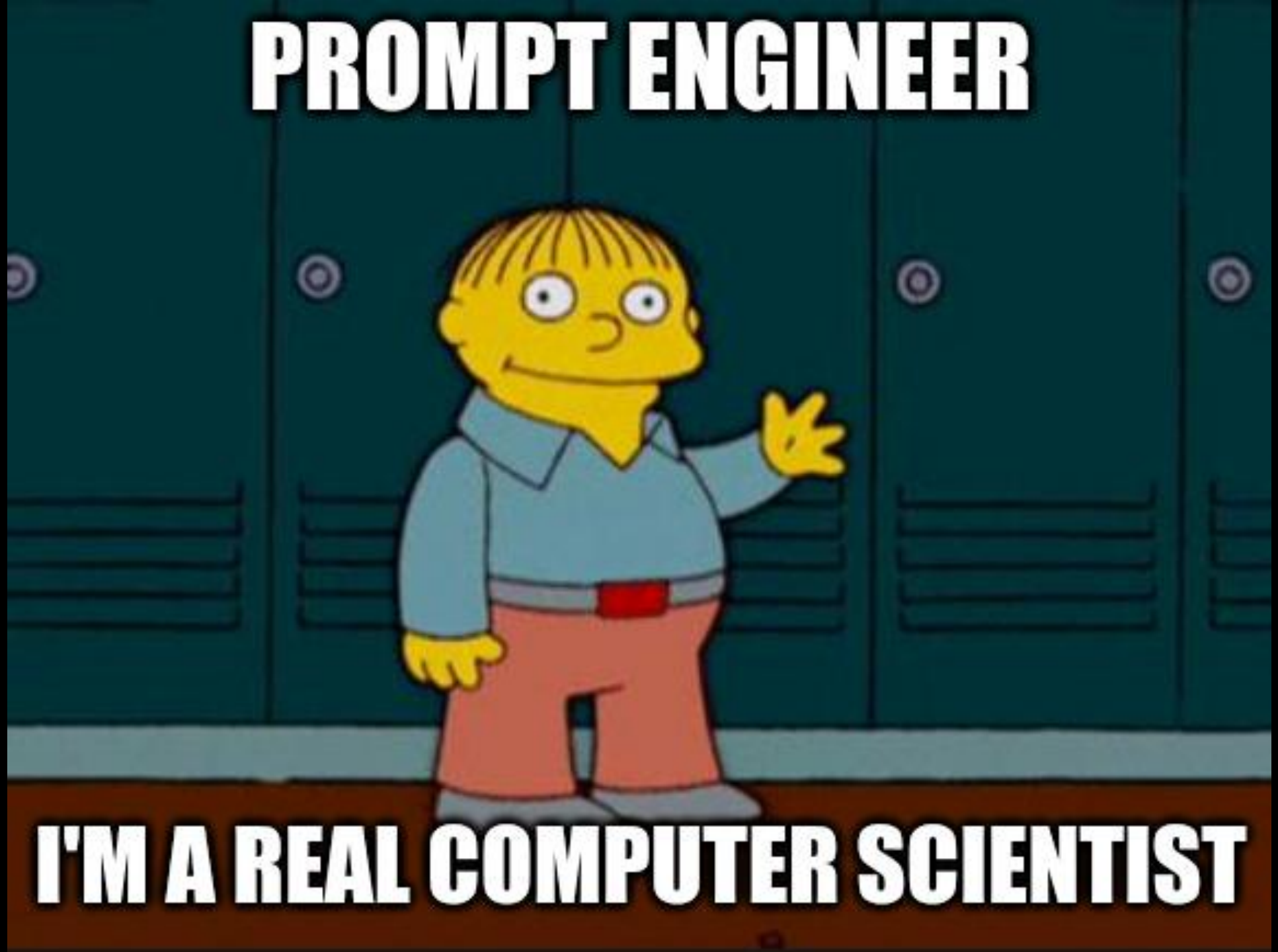
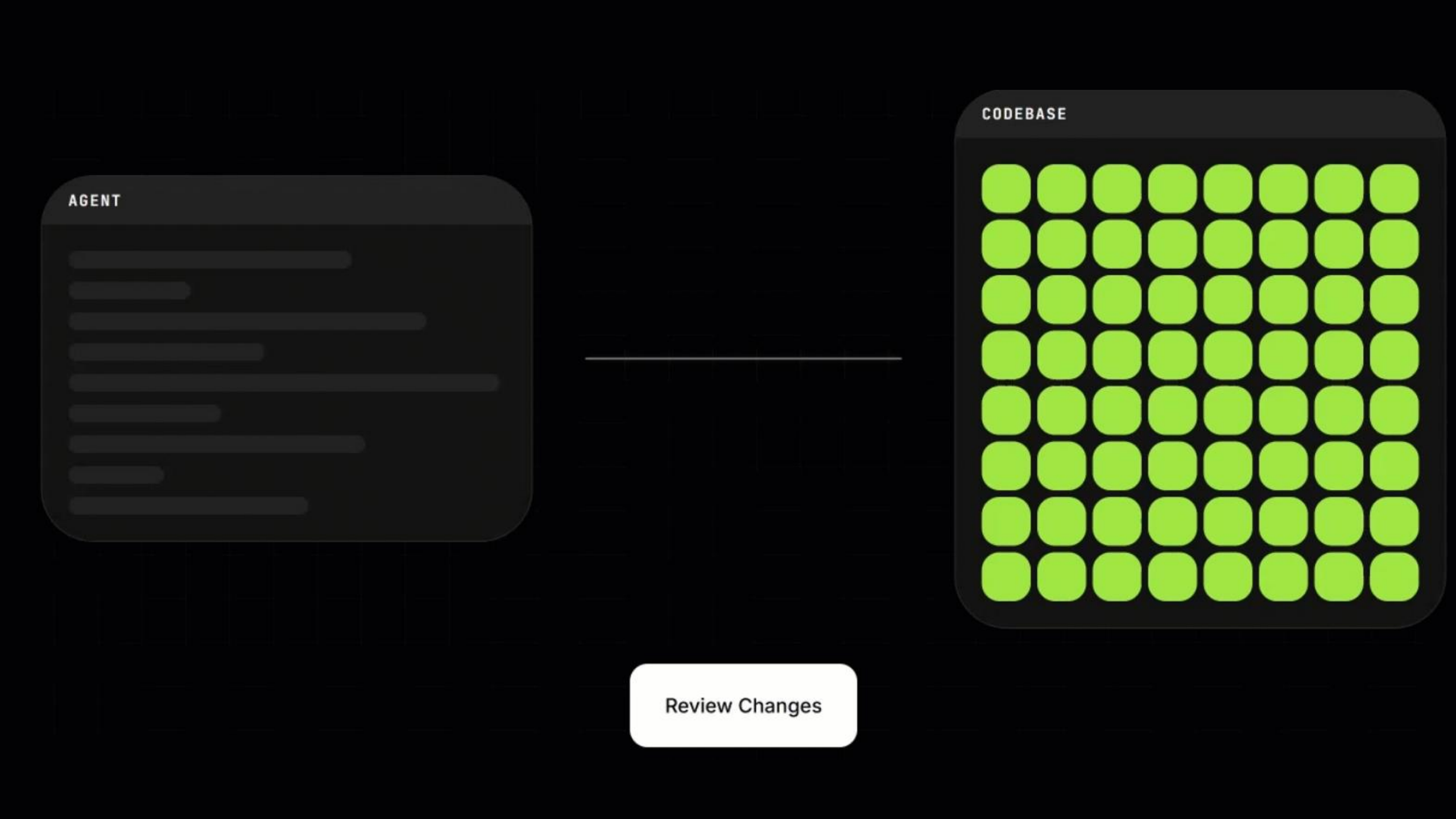
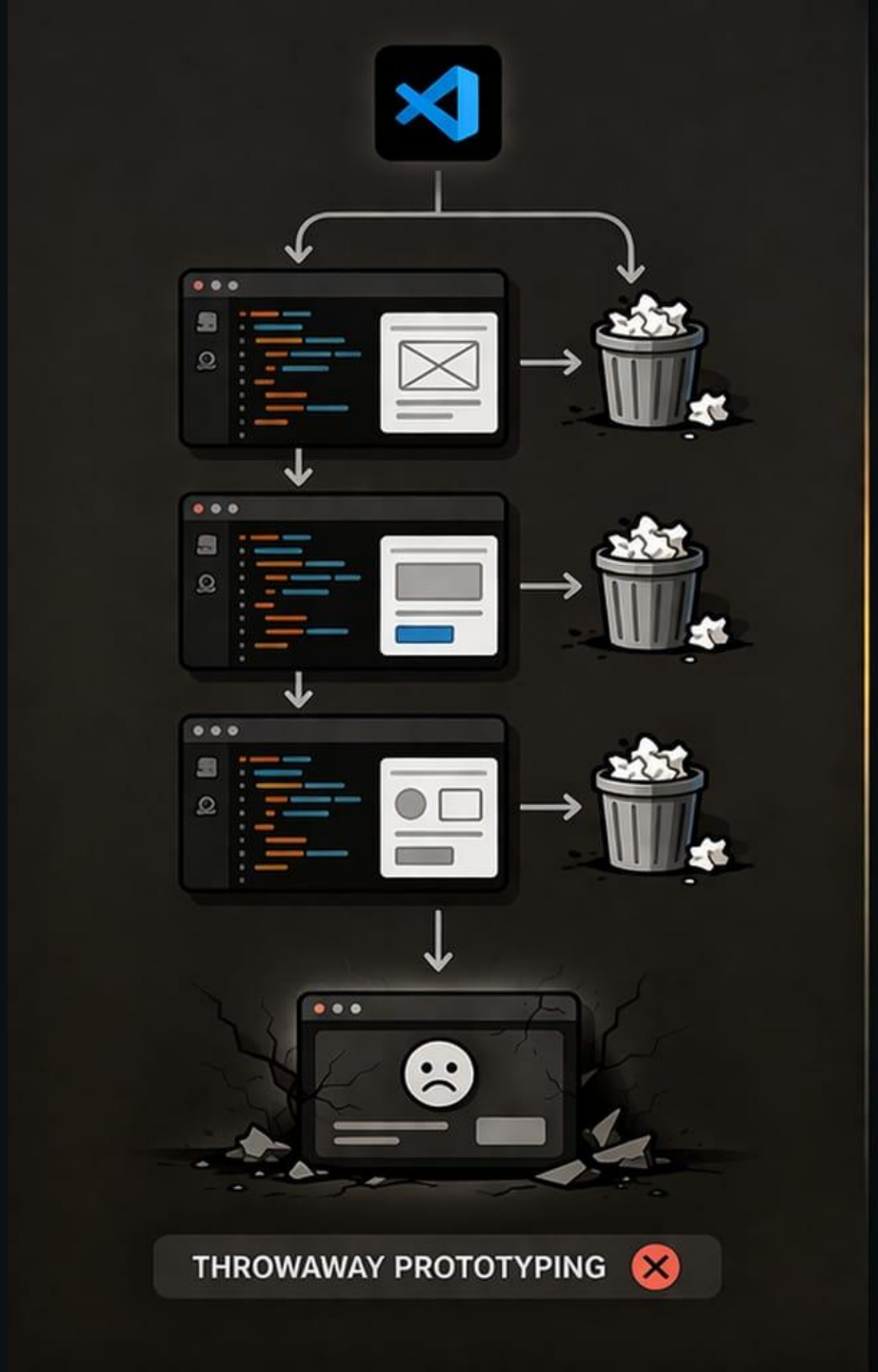

PROMPT ENGINEER
I'M A REAL COMPUTER SCIENTIST

VIBE CODING
SUPER
ACCEPT ALL
Pull Request #42
All checks passed
TO DO:
Ship it · test it · refactor - repeat


AGENT
CODEBASE
Review Changes


THROWAWAY PROTOTYPING

v1
v2
v3
EVOLUTIONARY PROTOTYPING

The Vibe Coding Workflow

Vibe code with structure, context, and control.

Ask Mode

Explore Before You Build

- Understand Codebase
- Explore Options
- Learn Patterns
- Research

Plan Mode

Shape the Solution

1. Analyze codebase
2. Identify files to change
3. Outline the approach
4. Refine with feedback
5. Ready to execute

Design first, code second

Agent Mode

Execute Across Files and Tools

Local Agent → Copilot CLI → Cloud Agent

copilot-instructions.md

Teach Copilot Your Conventions

- Tech Stack
- Coding Conventions
- API Design
- Error Handling
- Testing
- Workshop Context

→ Every response follows your rules

MCP Tooling

Extend Copilot with External Tools

mcp.json

```
{
  {"tools":
    {"filesystem", "github", "database"}}
}
```

Works in Ask, Plan, and Agent modes

What is an AI Agent?

Simple Chatbot

Ask question, get answer.
No memory, no actions.

AI Agent

Agent Profiles

Reusable Roles for Copilot

Select Agent

api-dev
Backend Expert

frontend-dev
UI Specialist

code-reviewer
Quality Guard

docs-writer
Documentation Writer

```
---
description: "A versatile coding assistant."
tools:
  - "code editor"
  - "terminal"
```

The Vibe Coding Workflow

Vibe Task Tracker

From Prompt to Production

Backend

- Java 21 / Spring Boot 4.0
- Task POJO + UUID
- ConcurrentHashMap store
- Constructor *injection*

API

- /api/tasks CRUD
- Filter + Sort
- Health + Favicon
- ResponseStatusException

Frontend

- Single index.html
- Dark GitHub theme
- Filter tabs
- ARIA + XSS Safe

Tests

- JUnit 5 + MockMvc
- 7 API Tests
- Context Smoke Test

Copilot Config

- copilot-instructions.md
- 4 Agent Profiles
- api-dev / frontend-dev
- code-reviewer / *docs-writer*

One prompt. Full-stack app. Copilot-ready.

BUILDING A VIBE TASK TRACKER WITH GITHUB COPILOT

in a new workspace, Build a **Vibe Task Tracker** — a workshop project showing how GitHub Copilot turns an idea into a practical workflow: Ask mode to explore, Plan mode to shape the solution, and Agent mode to execute across files and tools. We'll also cover `copilot-instructions.md`, reusable `.agent.md` profiles, and MCP tooling — so you can move fast, stay grounded, and vibe code with more structure, context, and control.

Backend: Java 21 / Spring Boot 4.0 / Maven (include wrapper). `Task` POJO with UUID string `id`, `title`, `description` (optional, default `""`), `completed` (default false), `createdAt` / `updatedAt` (Instant). Store tasks in a `@Repository` backed by `ConcurrentHashMap` — no database. Use constructor injection everywhere, `ResponseStatusException` for errors, and a `@RestControllerAdvice` `GlobalExceptionHandler` that returns `{ "error": "..." }` and never exposes stack traces.

Todos (13/14)

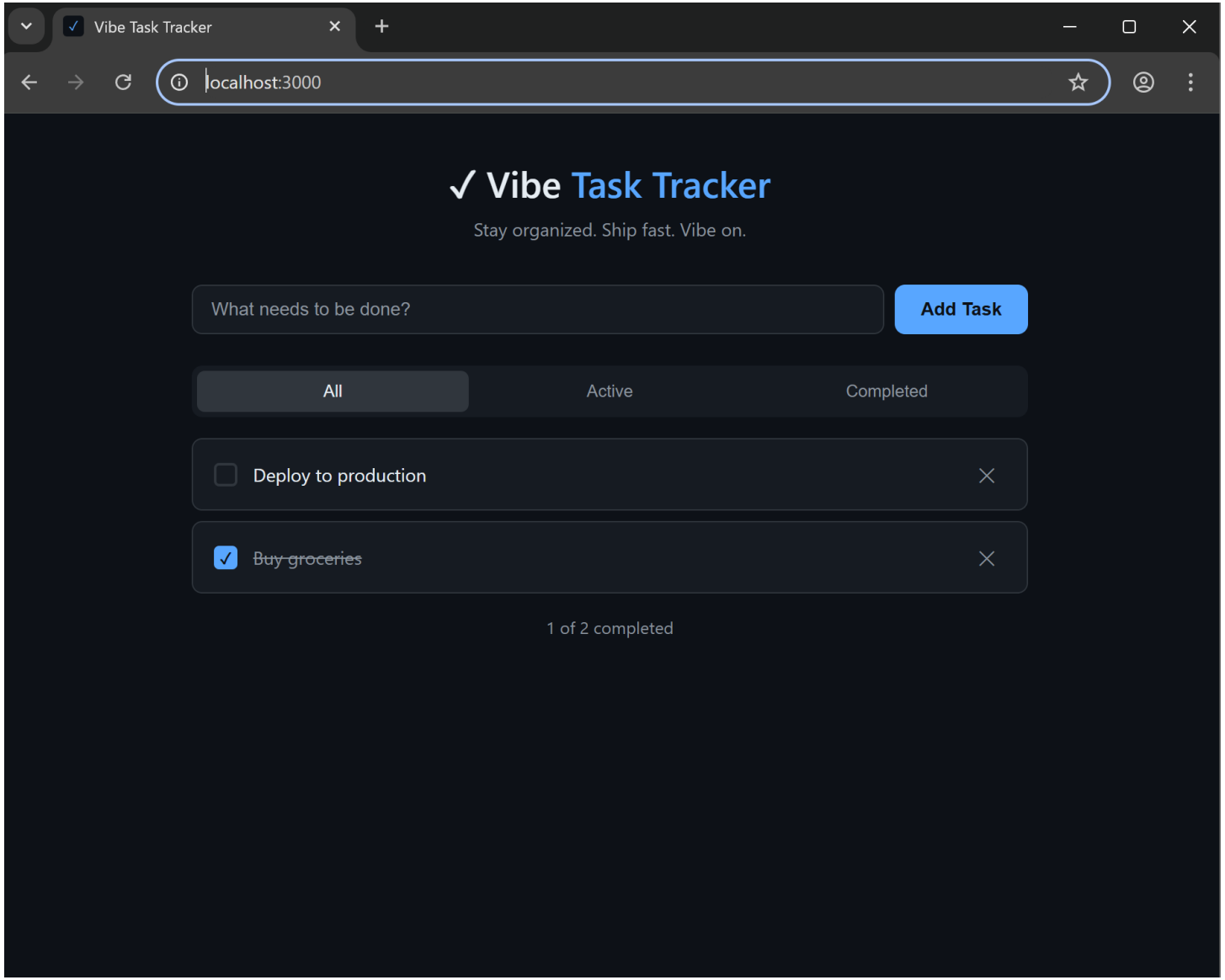
- Create application.properties
- Create frontend index.html
- Create JUnit 5 tests
- Create copilot-instructions.md
- Create agent.md profiles
- Build project & run tests

15 files changed +1083 -15 Keep Undo

Describe what to build

Agent Claude Opus 4.6 (1M context)(Internal only) · High

Local Default Approvals

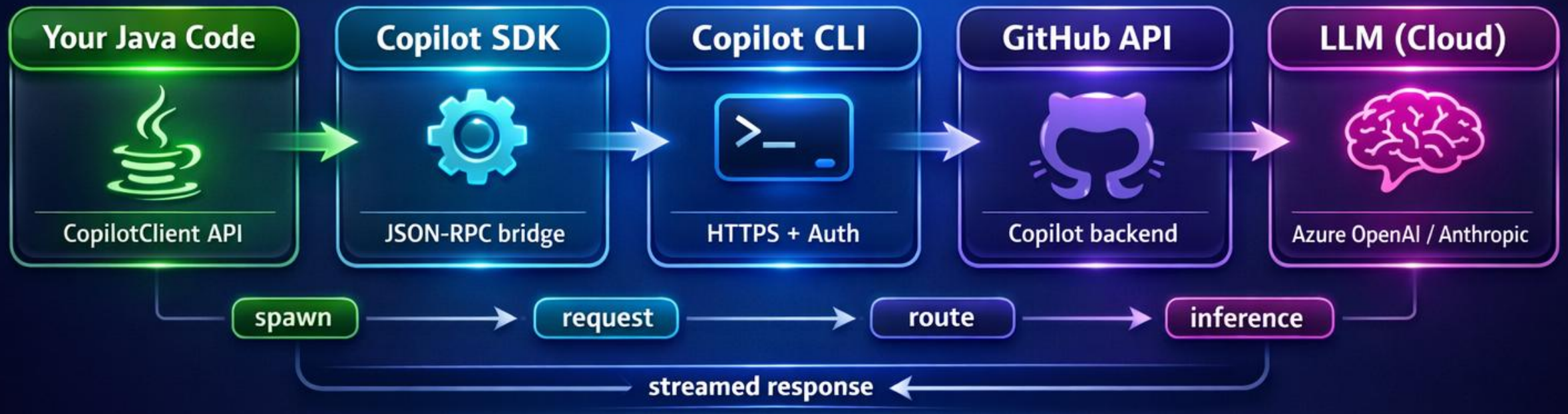


https://github.com/roryp/vibecoding

https://github.github.com/copilot-sdk-java

Vibe Coding with Copilot SDK for Java
Connect to CopilotClient()
Create a session and select the LLM
Claude-sonnet-4.5
Send a prompt
What is 2+2?
Print the result
Result: 4

Copilot SDK for Java — How It Works



What Can You Build?

Coding Assistants

Team conventions built in

PR Reviewers

Summarize & suggest

Copilot SDK

Test Generators

Pattern-aware tests

Doc Bots

Codebase-synced docs

Agentic Workflows

Reason, plan & execute

Advanced SDK Features

System Messages

Customize AI behavior

Code Review

Thread safety & null checks

Multi-turn Chat

Context-aware conversations

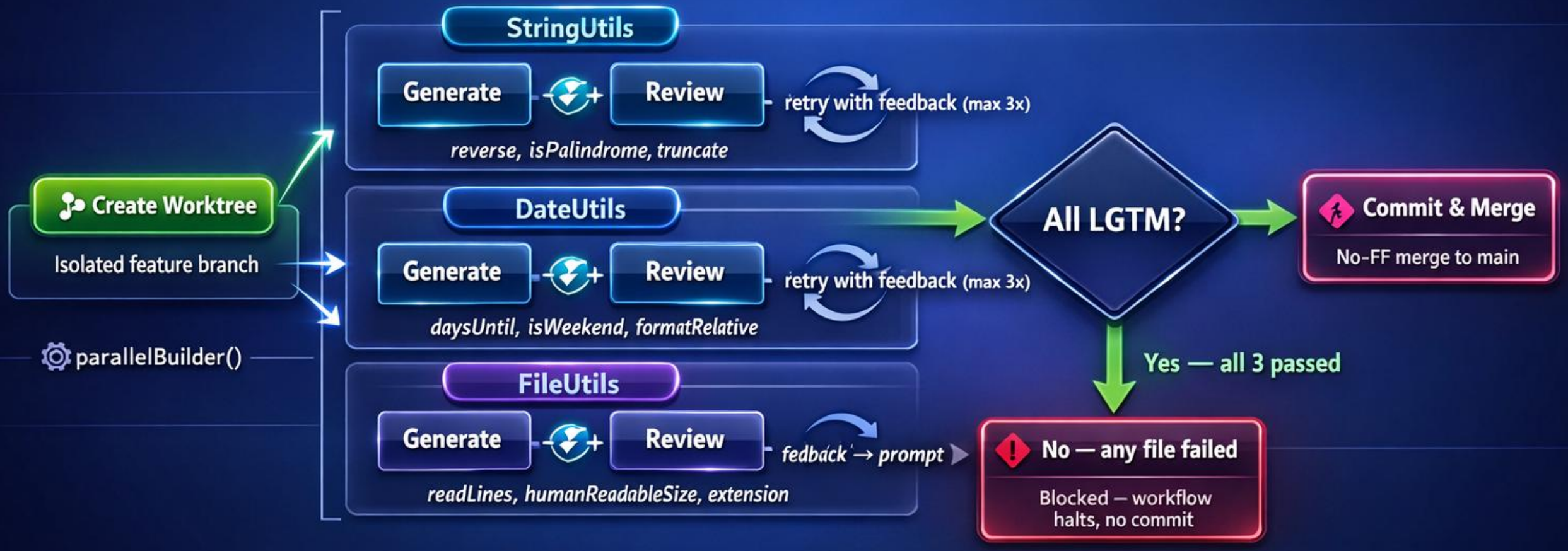
Structured JSON

Typed response parsing

Code Generation

Production-ready output

Parallel Generate → Review Workflow — LangChain4j + Copilot SDK



ALL files must pass review. If ANY one *fails* after 3 attempts, the entire merge is blocked.

https://github.com/roryp/github-copilot-sdk